Symbolic Infinitesimal Flexes for a Volume-Rigidity Conjecture in All Dimensions $d \geq 7$

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Abstract

Cruickshank, Jackson, and Tanigawa conjectured that if $\mathcal{S}$ is a simplicial 2-sphere on $\lceil d/2 \rceil + 2$ vertices and Z is a disjoint set of $d-3$ vertices, then the hypergraph of $(d-2)$-faces of the simplicial cone $\mathcal{S} * Z$ is volume rigid in $\mathbb{R}^d$. We give a symbolic counterexample for every $d \geq 7$. Put $m = d - 4$ and choose one distinguished cone vertex z_0. For a generic placement, let D be the $d \times m$ matrix of cone-vertex differences $p(z_i) - p(z_0)$. A nonzero symmetric matrix $K \in \mathrm{Sym}_m$ with zero diagonal, $1^\mathsf{T} K 1 = 0$, and $\mathrm{tr}(KD^\mathsf{T}D) = 0$ gives a symmetric affine motion $u(v) = DKD^\mathsf{T}(p(v) - p(z_0))$. The first two equations kill the facet contributions from the cone simplex, and the trace equation kills the hyperedges containing all cone vertices. Hence every squared $(d-2)$-volume derivative vanishes, while the motion is not Euclidean-trivial. Therefore $H_{d-2}(\mathcal{S} * Z)$ is not volume rigid in $\mathbb{R}^d$ for any $d \geq 7$. Finite modular computations for $4 \leq d \leq 9$ are reported only as cross-checks; the note does not claim an exact generic rank or exact generic corank formula.

1 Introduction

Cruickshank, Jackson, and Tanigawa introduced a framework for volume rigidity of uniform hypergraphs and simplicial complexes [CJT26]. Among their proposed base cases for higher-dimensional results is the following statement [CJT26, Conjecture 22]: for $d \geq 4$, if $\mathcal{S}$ is a simplicial 2-sphere on $\lceil d/2 \rceil + 2$ vertices and Z is a disjoint set of $d - 3$ vertices, then the hypergraph $H_{d-2}(\mathcal{S} * Z)$ is volume rigid in $\mathbb{R}^d$.

This note gives a uniform symbolic obstruction for all dimensions $d \geq 7$. The construction depends only on the way the $(d-2)$-faces of $\mathcal{S} * Z$ meet the cone simplex on Z. It is therefore independent of the combinatorial type of the simplicial 2-sphere $\mathcal{S}$.

The lower dimensions are not re-proved here. The source paper proves the corresponding cases $d = 4, 5, 6$, and the theorem below shows that the proposed family fails in every remaining dimension. The result is an infinitesimal generic non-rigidity statement. It does not address global uniqueness, nongeneric placements, exact generic rank, exact generic corank, or the broader manifold conjecture discussed in [CJT26].

2 Definitions

2.1 Simplicial cones and their hypergraphs

For a simplicial complex $\mathcal{S}$ and a finite set Z disjoint from $V(\mathcal{S})$, define

$$\mathcal{S} * Z = \{X \cup Z' : X \in \mathcal{S},\ Z' \subseteq Z\}.$$

For a simplicial complex $\mathcal{K}$, the hypergraph $H_k(\mathcal{K})$ has vertex set $V(\mathcal{K})$ and one hyperedge for every k-face of $\mathcal{K}$; hence every hyperedge has $k + 1$ vertices.

Fix $d \geq 7$. Let $\mathcal{S}$ be a simplicial 2-sphere on

$$s = \left\lceil \frac{d}{2} \right\rceil + 2$$

vertices, and let

$$Z = \{z_0, z_1, \dots, z_m\}, \qquad m = d-4,$$

be disjoint from $V(\mathcal{S})$. Thus $|Z| = m+1 = d-3$. The $(d-1)$-element hyperedges of $H_{d-2}(\mathcal{S} * Z)$ are exactly

$$e \cup Z \quad (e \in \mathcal{S}^{(1)}), \qquad t \cup \big(Z \setminus \{z_i\}\big) \quad (t \in \mathcal{S}^{(2)},\ 0 \le i \le m). \tag{1}$$

Indeed, such a face contains either two sphere vertices and all $m+1 = d-3$ cone vertices, or three sphere vertices and $m = d-4$ cone vertices.

2.2 Squared-volume Jacobian

Let $p : V(H_{d-2}(\mathcal{S} * Z)) \to \mathbb{R}^d$ be a placement. It is *generic* if its scalar coordinates are algebraically independent over $\mathbb{Q}$. For a hyperedge $\Delta = \{v_0, v_1, \dots, v_{d-2}\}$, let $M_\Delta(p)$ be the $(d-2) \times d$ matrix whose i-th row is $(p(v_i) - p(v_0))^\mathsf{T}$, and put

$$g_\Delta(p) = \det\big(M_\Delta(p) M_\Delta(p)^\mathsf{T}\big).$$

Then

$$g_\Delta(p) = ((d-2)!)^2 \operatorname{Vol}_{d-2}(p(\Delta))^2.$$

Let $J_{\mathcal{S},d}(p)$ be the squared-volume Jacobian with one row $\nabla g_\Delta(p)$ for each hyperedge Δ. At a generic placement this Jacobian has the same rank and right kernel as the volume-rigidity matrix of [CJT26]: the Gram determinant differs from squared volume by a fixed nonzero factor, and the row normalizing factors used in that paper are nonzero at a generic placement.

The Euclidean-trivial infinitesimal motions are

$$u(v) = Ap(v) + t, \qquad A^\mathsf{T} = -A, \quad t \in \mathbb{R}^d.$$

For a generic placement with at least $d+1$ vertices, these form a

$$d + \binom{d}{2} = \binom{d+1}{2}$$

dimensional subspace of $\ker J_{\mathcal{S},d}(p)$. The generic placement here has

$$|V(\mathcal{S})| + |Z| = \left\lceil \frac{d}{2} \right\rceil + d - 1 \ge d+1$$

vertices, so affine spanning is generic.

3 Main result

Theorem 3.1. *Let $d \ge 7$, let $\mathcal{S}$ be any simplicial 2-sphere on $\lceil d/2 \rceil + 2$ vertices, let Z be a disjoint $(d-3)$-element set, and let*

$$p : V(H_{d-2}(\mathcal{S} * Z)) \to \mathbb{R}^d$$

be generic. Then $J_{\mathcal{S},d}(p)$ has a non-Euclidean infinitesimal motion in its kernel. Consequently $H_{d-2}(\mathcal{S} * Z)$ is not volume rigid in $\mathbb{R}^d$, and Conjecture 22 of [CJT26] is false for every $d \ge 7$.

The remainder of this section constructs the motion and checks all hyperedge derivatives. The equivalence between generic volume rigidity and generic infinitesimal volume rigidity is Proposition 6 of [CJT26].

3.1 A trace formula for squared volume

For a linear subspace $U \subseteq \mathbb{R}^d$, write P_U for its orthogonal projector.

Lemma 3.2. *Let Δ be a nondegenerate $(d-2)$-simplex with direction space U, and let B be a symmetric $d \times d$ matrix. Under the affine velocity field $u(q) = B(q - q_0)$, where q_0 is fixed,*

$$\left.\frac{d}{dt}\right|_{t=0} g_\Delta\big(q_0 + (I + tB)(p - q_0)\big) = 2 g_\Delta(p)\,\mathrm{tr}(P_U B).$$

The same formula holds with g_Δ replaced by squared $(d-2)$-volume.

Proof. Translations do not affect edge differences. If $M = M_\Delta(p)$, then the transformed difference matrix is $M_t = M(I + tB)^\mathsf{T}$. Since $B = B^\mathsf{T}$,

$$\left.\frac{d}{dt}\right|_{t=0} M_t M_t^\mathsf{T} = 2 M B M^\mathsf{T}.$$

Jacobi's determinant identity gives

$$\begin{aligned}\left.\frac{d}{dt}\right|_{t=0} \det(M_t M_t^\mathsf{T}) &= 2\det(MM^\mathsf{T})\,\mathrm{tr}\Big((MM^\mathsf{T})^{-1} M B M^\mathsf{T}\Big) \\ &= 2 g_\Delta(p)\,\mathrm{tr}\Big(M^\mathsf{T}(MM^\mathsf{T})^{-1} M B\Big).\end{aligned}$$

The matrix $M^\mathsf{T}(MM^\mathsf{T})^{-1}M$ is P_U, which proves (3.2). Division by $((d-2)!)^2$ gives the squared-volume version. □

3.2 The auxiliary matrix

Translate notation so that $p(z_0)$ is the origin, and define

$$D = \begin{bmatrix} d_1 & d_2 & \cdots & d_m \end{bmatrix} = \begin{bmatrix} p(z_1) - p(z_0) & p(z_2) - p(z_0) & \cdots & p(z_m) - p(z_0) \end{bmatrix}, \qquad G = D^\mathsf{T} D.$$

For generic p, the $d \times m$ matrix D has full column rank.

Let $\mathcal{K}_G$ be the vector space of symmetric $m \times m$ matrices satisfying

$$K_{ii} = 0 \quad (1 \le i \le m), \qquad 1^\mathsf{T} K 1 = 0, \qquad \mathrm{tr}(KG) = 0. \tag{2}$$

The first condition leaves $\binom{m}{2}$ off-diagonal variables. The last two conditions are two homogeneous linear equations in those variables. For generic G they are independent: $1^\mathsf{T} K 1$ has coefficient vector proportional to the all-ones vector on off-diagonal entries, while $\mathrm{tr}(KG)$ has coefficient vector proportional to the off-diagonal entries of G, and these entries are not generically all equal. Hence

$$\dim \mathcal{K}_G = \binom{m}{2} - 2.$$

Since $m = d - 4 \ge 3$, this dimension is positive. Choose any nonzero $K \in \mathcal{K}_G$, and put

$$B = DKD^\mathsf{T}, \qquad u(v) = B\big(p(v) - p(z_0)\big). \tag{3}$$

Then B is symmetric and nonzero. Indeed, if $DKD^\mathsf{T} = 0$ and D has full column rank, then multiplying by a left inverse of D and its transpose gives $K = 0$.

Remark 3.3. The dimension $\binom{m}{2} - 2$ is used here only to guarantee that a nonzero K exists. This note does not claim that it equals the generic corank contribution.

3.3 Facet identities

Let $L = \operatorname{im} D$. Inside L, the cone simplex on Z has affine coordinates $0, e_1, \dots, e_m$. Its $m+1$ facet normal covectors are

$$e_1, \dots, e_m, \quad 1 = (1, \dots, 1)^\mathsf{T}.$$

The zero diagonal and $1^\mathsf{T} K 1 = 0$ therefore give

$$h^\mathsf{T} K h = 0 \quad \text{for } h \in \{e_1, \dots, e_m, 1\}. \tag{4}$$

The trace equation gives

$$\operatorname{tr}(B) = \operatorname{tr}(DKD^\mathsf{T}) = \operatorname{tr}(KG) = 0. \tag{5}$$

Also B is supported on L: it kills $L^\perp$ and maps $\mathbb{R}^d$ into L.

3.4 Stationarity of all hyperedge volumes

First consider a hyperedge $\Delta = e \cup Z$. Its direction space has the orthogonal decomposition

$$U_\Delta = L \oplus R,$$

where $R \subseteq L^\perp$ is the span of the projections of the two sphere vertices in e. Therefore

$$\operatorname{tr}(P_{U_\Delta} B) = \operatorname{tr}(P_L B) + \operatorname{tr}(P_R B) = \operatorname{tr}(B) = 0. \tag{6}$$

Now let

$$\Delta = t \cup (Z \setminus \{z_\ell\}).$$

Denote by L_ℓ the $(m-1)$-dimensional direction space of the corresponding facet of the cone simplex on Z. Let h_ℓ be its normal covector in the D-coordinates. Thus

$$h_\ell \in \{e_1, \dots, e_m, 1\}.$$

The vector

$$n_\ell = DG^{-1} h_\ell$$

is normal to L_ℓ inside L, because $D^\mathsf{T} n_\ell = h_\ell$. From (4),

$$n_\ell^\mathsf{T} B n_\ell = h_\ell^\mathsf{T} K h_\ell = 0. \tag{7}$$

Since $L = L_\ell \oplus \operatorname{span}(n_\ell)$ orthogonally, (5) and (7) imply

$$\operatorname{tr}(P_{L_\ell} B) = 0.$$

The direction space of Δ decomposes as

$$U_\Delta = L_\ell \oplus R_\ell$$

for a subspace $R_\ell \subseteq L_\ell^\perp$. Moreover,

$$L_\ell^\perp = \operatorname{span}(n_\ell) \oplus L^\perp.$$

For $w = \lambda n_\ell + r$, with $r \in L^\perp$, the facts $Br = 0$ and (7) give $w^\mathsf{T} B w = 0$. Since B is symmetric, polarization shows that the compression of B to $L_\ell^\perp$ is zero. Hence $\operatorname{tr}(P_{R_\ell} B) = 0$, and altogether

$$\operatorname{tr}(P_{U_\Delta} B) = 0. \tag{8}$$

Equations (6) and (8) cover the two hyperedge families in (1). By Lemma 3.2, the velocity u in (3) lies in the kernel of the generic squared-volume Jacobian.

3.5 Nontriviality

Suppose, for contradiction, that u is Euclidean-trivial. Then there are a skew-symmetric matrix A and a vector t such that

$$B(p(v) - p(z_0)) = Ap(v) + t$$

for every vertex v. Substitution of $v = z_0$ gives $t = -Ap(z_0)$, so

$$B(p(v) - p(z_0)) = A(p(v) - p(z_0))$$

for every vertex v. A generic placement affinely spans $\mathbb{R}^d$; its differences from $p(z_0)$ therefore span $\mathbb{R}^d$, forcing $B = A$. This is impossible because B is nonzero and symmetric while A is skew-symmetric. Thus the constructed kernel vector is not Euclidean-trivial, proving Theorem 3.1.

4 The dimension-seven specialization

When $d = 7$, one has $m = 3$, $|Z| = 4$, and $|V(\mathcal{S})| = 6$. Write

$$G = D^{\mathsf{T}}D, \qquad a = G_{12}, \quad b = G_{13}, \quad c = G_{23}.$$

Then $\mathcal{K}_G$ is one-dimensional and may be generated by

$$\xi = c - b, \qquad \eta = a - c, \qquad \zeta = b - a, \qquad K = \begin{pmatrix} 0 & \xi & \eta \\ \xi & 0 & \zeta \\ \eta & \zeta & 0 \end{pmatrix}.$$

Indeed,

$$\xi + \eta + \zeta = 0, \qquad a\xi + b\eta + c\zeta = 0,$$

which are exactly the equations $1^{\mathsf{T}}K1 = 0$ and $\operatorname{tr}(KG) = 0$.

Every six-vertex simplicial 2-sphere has twelve edges and eight triangles. Thus $H_5(\mathcal{S} * Z)$ has

$$12 + 4 \cdot 8 = 44$$

hyperedges. A generic ten-point placement in $\mathbb{R}^7$ has 70 velocity coordinates and 28 Euclidean-trivial motions, so rigidity would require rank 42. The construction above gives at least one additional nontrivial kernel vector, hence the generic rank is at most 41.

5 Finite cross-checks and verification boundary

The proof of Theorem 3.1 is the symbolic argument in the preceding sections. The following finite computations are included only as independent cross-checks of the hyperedge construction and derivative matrices. They do not certify exact generic rank or exact generic corank.

A direct modular Jacobian checker reconstructed the relevant simplicial 2-sphere types from the graph atlas, formed all hyperedges in (1), and differentiated the Gram determinants over three large primes and three random seeds. The observed ranks were stable in the following tests.

d	sphere types tested	rows	columns	observed rank
4	1	10	20	10
5	1	21	35	20
6	1	27	48	27
7	2	44	70	40
8	2	52	88	47
9	5	75	117	63

A separate affine-flex-space checker solved the equations $\mathrm{tr}(P_U B) = 0$ for symmetric affine matrices B at the same specialized placements. Its observed dimensions were $0, 0, 0, 2, 5, 9$ for $d = 4, 5, 6, 7, 8, 9$, respectively. For $d \geq 7$, this is consistent with the symbolic construction and with additional affine motions in those finite specializations. The public theorem uses only the existence of one such non-Euclidean motion for generic placements.

The $d = 7$ package also contains the older exact hyperedge tables for the two six-vertex sphere types. They are retained for reproducibility and link stability.

6 Relation to the source paper and scope

The source paper proves the lower-dimensional range $d = 4, 5, 6$ for the relevant base cases and states Conjecture 22 as a proposed family for extending the method [CJT26]. The theorem here disproves that conjecture in every dimension $d \geq 7$. It does not settle the source paper's broader Conjecture 2, since that extension involves additional manifold-rigidity arguments beyond this base-case family.

This note makes no claim of public priority, of being the first counterexample, or of establishing the current public status of the all-dimensional conjecture. Public novelty or priority is `NOT_ESTABLISHED`.

A Dimension-seven hyperedge lists

In the tables below, a string such as `012678` denotes the six-element set $\{0, 1, 2, 6, 7, 8\}$. Each table contains all 44 hyperedges in the order used for the older exact $d = 7$ cross-check.

Stacked type

012678	012679	012689	012789
013678	013679	013689	013789
016789	023678	023679	023689
023789	026789	036789	124678
124679	124689	124789	126789
134678	134679	134689	134789
136789	146789	235678	235679
235689	235789	236789	245678
245679	245689	245789	246789
256789	345678	345679	345689
345789	346789	356789	456789

Octahedral type

012678	012679	012689	012789
013678	013679	013689	013789
016789	024678	024679	024689
024789	026789	034678	034679
034689	034789	036789	046789
125678	125679	125689	125789
126789	135678	135679	135689
135789	136789	156789	245678
245679	245689	245789	246789
256789	345678	345679	345689
345789	346789	356789	456789

References

[CJT26] James Cruickshank, Bill Jackson, and Shin-ichi Tanigawa. Volume rigidity of simplicial manifolds. *Combinatorica*, 46:23, 2026.